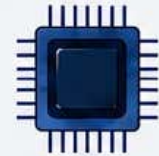
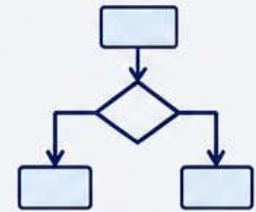




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Syllabus



Minor Course: Software Engineering and Java Programming

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	112267	Software Engineering and Java Programming	2	30	2 Hrs.	30

Course Objectives:	1. Understand the fundamental concepts of Software Engineering 2. Analyse software requirements, risk management techniques, and software quality measurement methods used in software development. 3. . standards for producing reliable software products.			
Course Outcomes:	Students will be able to - 1. Understand the basic principles of Software Engineering 2. Identify and analyse functional and non-functional requirements using requirement engineering techniques. 3. To understand the concepts and features of Java Programming. 4. The skills to apply Decision making and Branching in Java programming			
Unit System	Contents	Workload Allotted	Weightage of Marks Allotted	Incorporation of Pedagogies
Unit I	Introduction to Software Engineering: Characteristics of Software, Types of Software, Need of Software Engineering, Software Development Life Cycle (SDLC), Phases of SDLC, Software Development Models: Introduction to Process Models, Waterfall Model.	8 Hrs.	8 Marks	Use any pedagogical technique suitable for the topic
Unit II	Software Requirements: Functional and non-functional requirements, User requirements, System requirements Requirements engineering process: Requirements elicitation and analysis, Requirements validation, Requirements management. Metrics for Process and Products: Software Measurement, Metrics for software quality	7 Hrs.	7 Marks	
Unit III	Introduction to JAVA: History and evolution, Feature, JRE, JDK, JVM, Data types, Keywords, Operators, Structure of Java Program, Java Command line argument, Simple java programs.	8 Hrs.	8 Marks	
Unit IV	Decision Making and Branching: Simple if statement, if...else statement, Nesting of if...else statement, Else...if ladder, switch statement. Decision Making and Looping: While statement, Do-While statement, For statement, jumps in loop	7 Hrs.	7 Marks	
References:	Course Material/Learning Resources			

Text books:

5. Software Engineering: A Practitioner's Approach – Roger S. Pressman
6. Software Engineering – Ian Sommerville
7. Programming with Java A Primer, Fourth Edition- E. Balguruswami (McGraw Hill)

Reference Books:

1. Software Engineering Principles and Practice – Hans Van Vliet
2. Object Oriented and Classical Software Engineering – Stephen R. Schach
3. Software Engineering – K. K. Aggarwal & Yogesh Singh
4. Java - The Complete Reference 11th edition - Herbert Schildt (McGraw Hill)
5. Introduction to Java programming 6th edition, Y. Daniel Liang, Pearson education.
6. "Java Handbook" Patricks Naughton, Osborne McGraw Hill

Weblink to Equivalent MOOC on SWAYAM if relevant:

1. https://onlinecourses.nptel.ac.in/e-learning/preview/noc25_cs108
2. <https://cse.iitk.ac.in/pages/CS455.html>
3. https://onlinecourses.swayam2.ac.in/aic20_sp13/preview
4. https://onlinecourses.nptel.ac.in/noc22_cs47/preview
5. https://onlinecourses.nptel.ac.in/e-learning/preview/noc23_cs85

Weblink to Equivalent Virtual Lab if relevant:

<https://se-iitkgp.vlabs.ac.in/>

Any pertinent media (recorded lectures, YouTube, etc.) if relevant:

1. <https://www.youtube.com/watch?v=Z6f9ckEElU>
2. https://onlinecourses-archive.nptel.ac.in/noc18_cs43/preview
3. https://www.youtube.com/playlist?list=PL_pbwdIyffslgxMVyXhnHiSn_EWTvx1G-
4. <https://www.youtube.com/watch?v=mglWM6PKZVk>
5. <https://www.youtube.com/watch?v=wBJDrRBPmuY>
6. <https://www.youtube.com/watch?v=Qg3HDmbW6I>

Assessment Rubric: Minor Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI THEORY EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Theory	Course Code: 112267	Course Title: Software Engineering and Java Programming	Max Marks: 20
Sr. No.	Assessment Criteria		Marks
1	Any Two of the Activity (weekly or monthly assignment/ Unit Test/Class Test) (CAT-1)		10
2	Any One of the Activity includes Open Book Test, MCQ Tests, / Theory related Class activity/Task etc.) etc. (CAT-2)		5
3	Participation in any one Activity includes Quiz, Seminar, Group Discussion, Industry visit, Case Study, Mini project, innovative work, etc. based on Theory Course. Involvement in subject related activities such as Conference, Workshop, Coding Competitions etc. (CAT-3)		5